Questions

Q1.

Some questions must be answered with a cross in a box (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

When water boils and turns into steam, there are changes in the arrangement of particles and the density.

Which of these shows the changes?

(1)

		space between particles in steam	density of steam
☐	A	bigger than in water	greater than water
☐	B	bigger than in water	less than water
☐	C	smaller than in water	greater than water
☐	D	smaller than in water	less than water

(Total for question = 1 mark)

Q2.

A student determines the volume of a piece of metal by measuring the volume of water that it displaces.

The student wrote the following in his notebook.

> I put some water into a measuring cylinder.
> I put the piece of metal into the water in the measuring cylinder.
> I took the reading of the new water level in the measuring cylinder.
> This was the volume of the piece of metal.

The student's description is incomplete.

Suggest **two** sentences that the student could have included to provide a more complete description of the correct procedure.

(2)

1 .....

.....

2 .....

.....

(Total for question = 2 marks)

Q3.

Figure 7 shows a shed made mostly of concrete blocks.

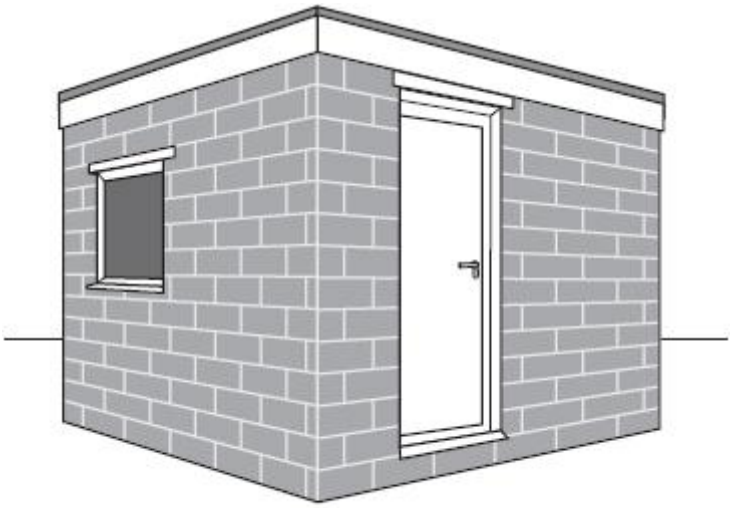

Figure 7

State **two** practical ways to reduce heat loss from this shed.

(2)

1 .....

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2 .....

.....

(Total for question = 2 marks)

Q4.

A student uses the apparatus in Figure 3 to determine the specific heat capacity of water.

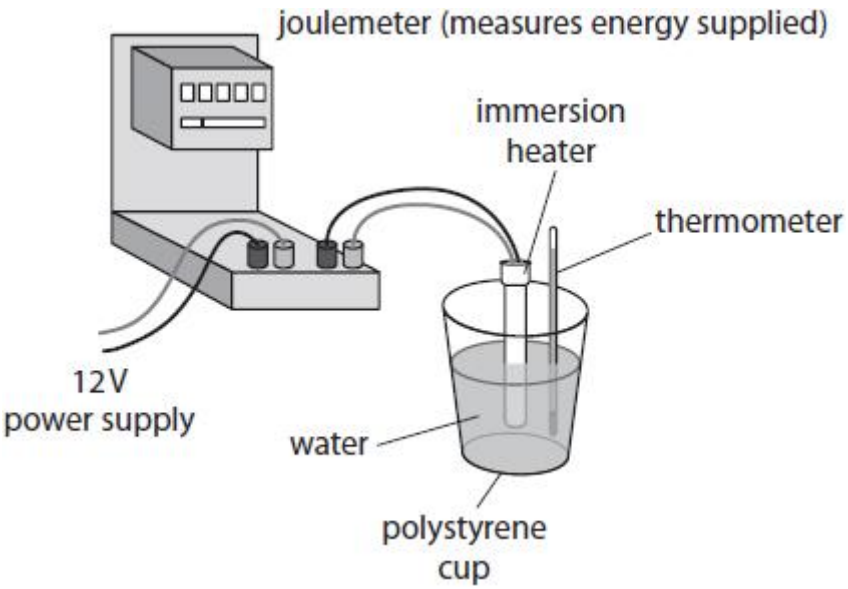

Figure 3

The student decides to measure the temperature of the water every minute while it is being heated.

Figure 4 shows a graph of the student's results.

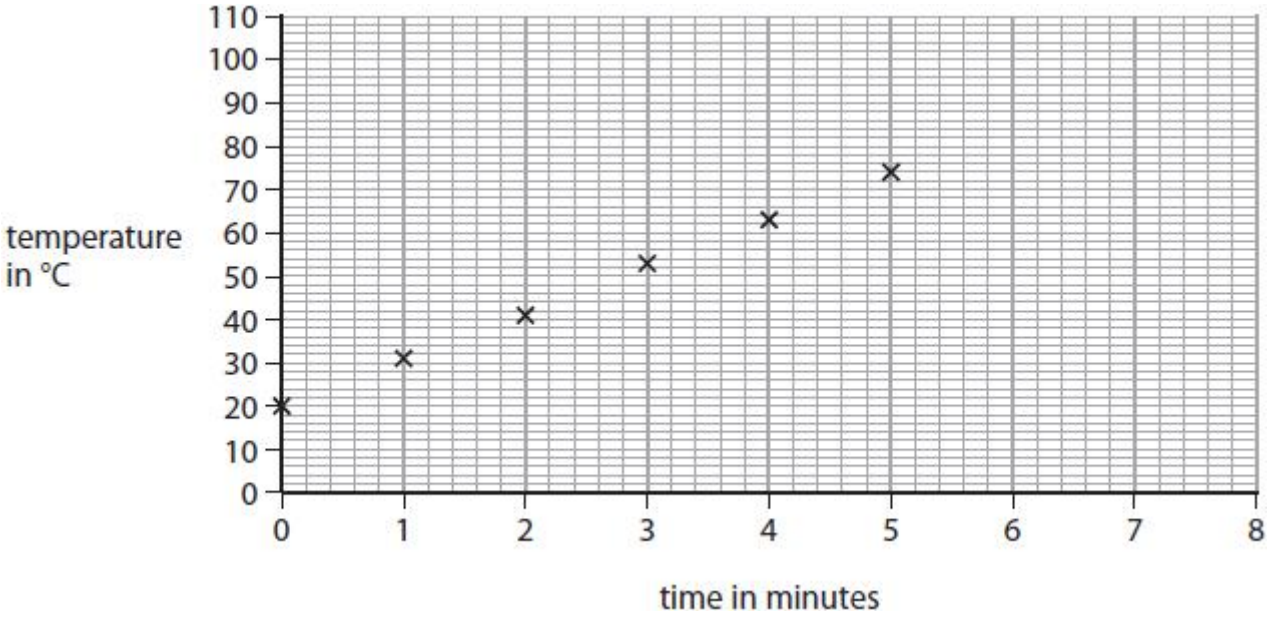

Figure 4

Predict the temperature of the water if the heating continues up to 8 minutes.

(1)

temperature of the water = ... °C

(Total for question = 1 mark)

Q5.

As part of the testing of different types of steel, a steelworker needs to obtain a temperature-time graph for **solidifying** molten steel.

Figure 9 shows an arrangement the steelworker could use.

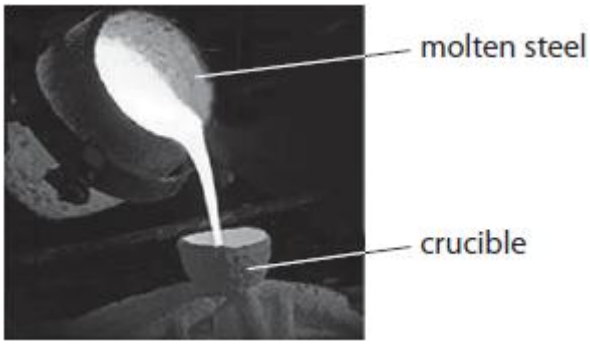

Figure 9

The following devices are available to the steel worker.
The melting point of these steels is between 1425 and 1540 °C

device	range of temperatures	other notes
Thermocouple thermometer	−50 to 1800 °C	Fast response time Probe inserted into melt
Infrared thermometer (pyrometer)	1200 to 2000 °C	Remotely read, using infrared radiation, measures the temperature of the surface it is aimed at
Platinum resistance thermometer	−200 to 850 °C	The most accurate of thermometers based on how resistance changes with temperature

Describe how the steelworker could obtain a temperature-time graph for steel as it goes from the liquid to the solid state.

(4)

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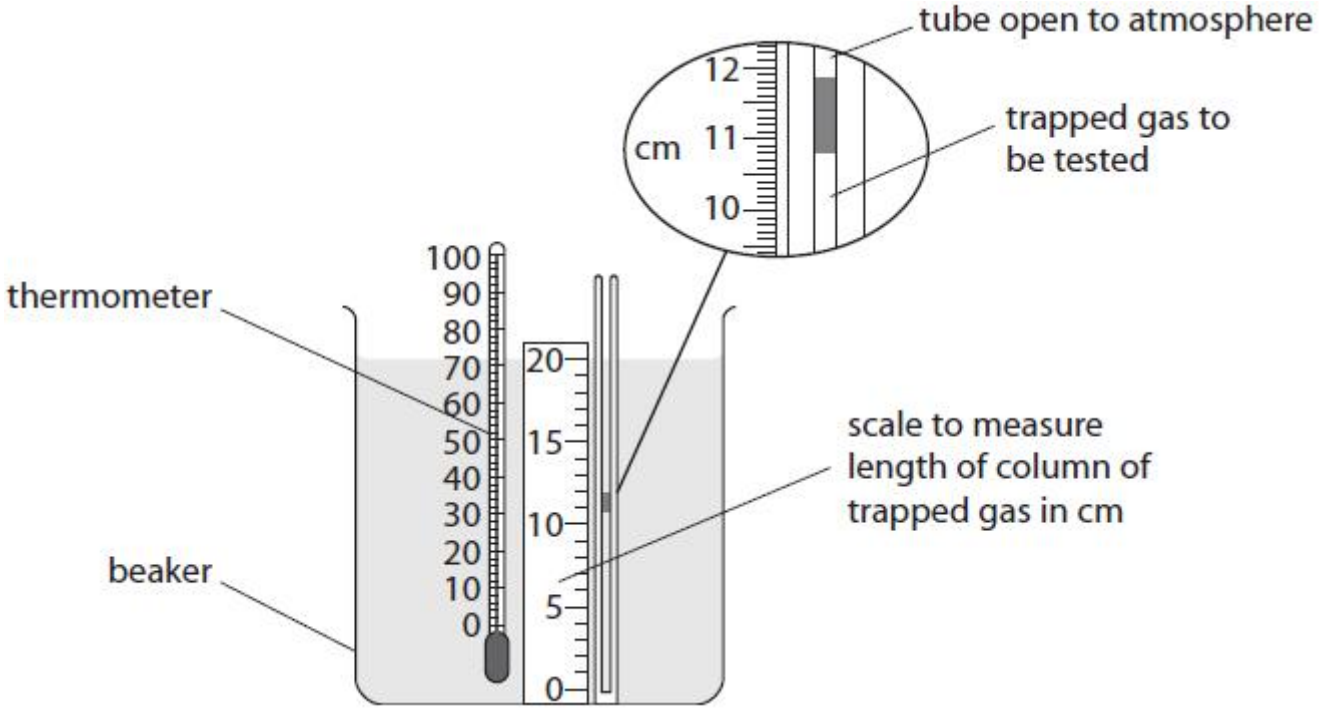
(Total for question = 4 marks)

Q6.

A student investigates how the volume of a gas changes when its temperature increases.

The diagram shows the equipment used and the length of the trapped gas at 25°C.



(a) (i) Use the scale to estimate the length of the column of trapped gas.

(1)

length of column of trapped gas = ... cm

(ii) Complete the sentence by putting a cross (☒) in the box next to your answer.
The cross-sectional area of the capillary tube is 1.94×10^{-3} cm^2.
The volume of the column of trapped gas at 25°C is about

(1)

☐ **A** $5.6 \times 10^{+3}$ cm^3
☐ **B** 2.1×10^{-2} cm^3
☐ **C** 2.1×10^{-3} cm^3
☐ **D** 5.6×10^{-4} cm^3

(iii) The gas is heated to 50°C.
The volume of the trapped gas at 50°C is 2.31×10^{-2} cm^3.
Calculate the volume of the trapped gas at 100°C.

(3)

volume of the trapped gas = ... cm^3

(b) Describe how the average kinetic energy of the particles of the gas changes as the temperature of the gas changes.

(3)

.....

.....

.....

.....

Q7.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these means changing state from solid directly to gas?

(1)

- ☐ **A** condensing
- ☐ **B** freezing
- ☐ **C** melting
- ☐ **D** sublimating

(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Figure 21 shows a sealed aerosol can.

Figure 21

The sealed can is taken from a cold room into a warm room.

If the volume of the can remains the same, which of these does **not** change?

- ☐ **A** the pressure inside the can
- ☐ **B** the mean speed of the particles inside the can
- ☐ **C** the mean distance between the particles inside the can
- ☐ **D** the mean size of the momentum of the particles inside the can

(1)

(Total for question = 1 mark)

Q9.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A teacher heats a block of wax until all the wax melts.

The teacher pours the melted wax into a container and allows the wax to cool.

Figure 7 shows how the temperature of the wax changes during a time of 20 minutes.

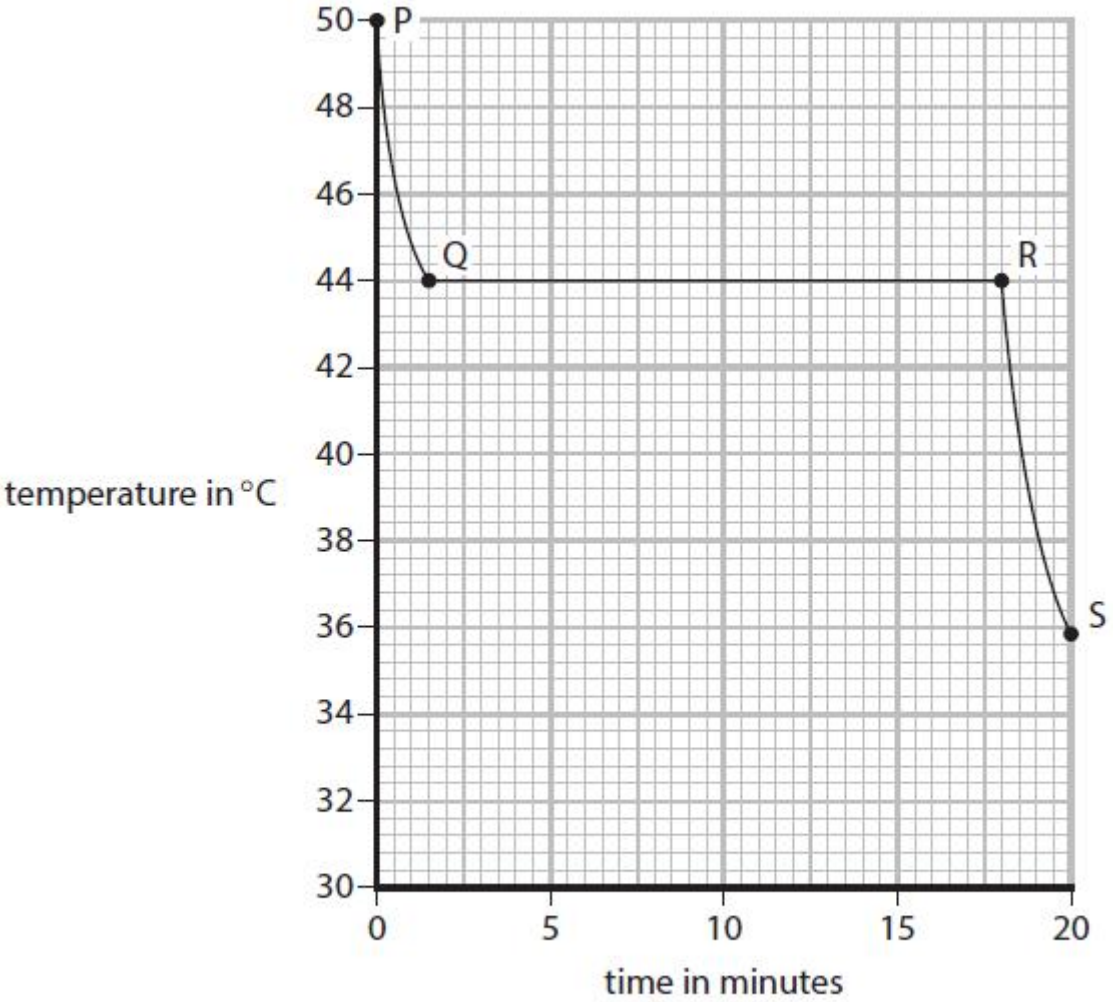


Figure 7

Four different points on the graph are labelled P, Q, R and S.

Between which two points is **all** the wax liquid?

(1)

- ☒ **A** P and Q
- ☒ **B** P and R
- ☒ **C** Q and S
- ☒ **D** R and S

(Total for question = 1 mark)

Mark Scheme

Q1.

Question number	Answer	Mark
	[x] **B** bigger than in water less than water A is incorrect because the density of steam is less than water. C is incorrect because the space between the particles increases. D is incorrect because the space between the particles increases and density of steam is less than water.	**(1)** **AO1.1**

Q2.

Question number	Answer	Additional guidance	Mark
	Any **two** from the following in any order	Answers need not be exactly the same as those given here provided that the meaning is clear.	**(2)** **AO1**
	(I took a) reading of the water level in the measuring cylinder without the metal. (1)	accept measured / read for take a reading accept reading of original level / volume accept starting with a specified amount e.g. 50ml	
	(I made sure that) the metal was fully immersed / submerged (1)	all the metal was under water	
	(I) subtracted the two readings / volumes (1)	took one from the other / found the difference ignore: repeat and/or average other measurements such as mass other methods such as Eureka can ideas of spillage reading from bottom of meniscus	

Q3.

Question number	Answer	Additional guidance	Mark
	statements to include any **two** from use cladding / (extra) insulation (1) use double thicknesses of the concrete (1) use silver / reflective / white (paint) (1) plant trees around (wind break) (1) use double glazed windows (1) (properly) close window(s)/door	 create cavity draft exclusion	**(2)** **AO1**

Q4.

Question Number:	Answer	Additional Guidance	Mark
	100 (°C) (1)	accept any answer between and including 95 and 102 (possibility that it is not pure water and possibility of heat loss prevents reaching boiling point)	**(1)** AO 2 1

Q5.

Question number	Answer	Additional guidance	Mark
	An answer that combines **any four** of the following points of understanding to provide a logical description: • chooses either thermocouple or infra-red thermometer (1) • molten steel is poured into a crucible (1) • a stopwatch is started (1) • the crucible + contents are allowed to cool down (in the room) (1) • temperatures are taken at regular intervals (e.g. every minute) (1)	any interval with steel – every 10 minutes etc.	**(4)**

Q6.

Question Number	Answer	Acceptable answers	Mark
(a) (i)	10.8 + or – 0.2 (cm)	Any value between 10.6(cm) and 11.0 (cm) Accept 11 cm	**(1)**

Question Number	Answer	Acceptable answers	Mark
(a)(ii)	B 2.1×10^{-2} cm^3		**(1)**

Question Number	Answer	Acceptable answers	Mark
(a)(iii)	Temperature conversion to K 50°C to 323K **OR** 100°C to 373K (1) Substitution $V_1 = \frac{2.31 \times 10^{-2} \times 373}{323}$ (1) Evaluation 2.67×10^{-2} (cm^3) (1)	 If equation is transformed to give V_2, allow correct substitution mark. 0.0267(cm^3), 2.7×10^{-2} (cm^3), 0.027(cm^3), 2.67×10^{-8} m^3, 2.7×10^{-8} m^3 Allow power of ten error for 2 marks e.g. 267 Allow 2.6×10^{-2} for 3 marks Full marks for correct answer with no working If temperature is not converted to Kelvin, maximum two marks e.g. $V_1 = \frac{2.31 \times 10^{-2} \times 100}{50}$ 4.62×10^{-2}(cm^3) Allow power of ten error for 1 mark e.g. 4.62 2 marks for 4.62×10^{-2} (cm^3) with no working	**(3)**

Question Number	Answer	Acceptable answers	Mark
(b)	A description including: (Average) KE/it increases as the temperature increases (1) Idea of proportionality / KE doubles when the temperature doubles (1) (when) temperature in Kelvin /K (1)	Allow energy for kinetic energy Or reverse argument • (Average) KE/it is (directly) proportional to the Kelvin temperature gets all three marks (Average) KE/it is (directly) proportional to the temperature gets first two marks Allow absolute scale	**(3)**

Q7.

Question number	**Answer**	**Mark**
	☒ D sublimating A is incorrect because it describes a change of state from gas to liquid. B is incorrect because it describes a change of state from liquid to solid C is incorrect because it describes a change of state from solid to liquid	**(1)** **AO1**

Q8.

Question Number	Answer	Mark
	The only correct answer is C the mean distance between the particles inside the can ***A, B and D have physical quantities which will all increase upon heating***	**(1)** **AO3.1**

Q9.

Question number	Answer	Additional guidance	Mark
	A P and Q B, C and D are incorrect because at least some of the wax is solid in each case.		**(1)** **AO3**